

University of Transport Ho Chi Minh City

FACULTY OF INFORMATION TECHNOLOGY

CAPSTONE PROJECT

**AI-POWERED PERSONALIZED
LEARNING PATH PLATFORM FOR
COMPETENCY ASSESSMENT EXAM
PREPARATION**

Project Title: AI-powered Personalized Learning Path Platform for Competency Assessment Exam Preparation

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1 Main Proposal Content

1.1 Context

In recent years, Competency Assessment Examinations have become one of the primary admission methods adopted by many universities in Vietnam.

Unlike traditional examinations that mainly evaluate memorized knowledge, competency assessment focuses on measuring students' logical reasoning, analytical thinking, reading comprehension, quantitative reasoning, and problem-solving abilities.

Although numerous online learning platforms and examination preparation systems are available, most of them provide identical learning content and predefined study plans for every learner.

Such an approach does not consider individual differences in academic background, learning pace, target scores, available study time, or knowledge mastery.

With the rapid advancement of Artificial Intelligence and Educational Data Analytics, personalized learning has become a promising approach for improving learning efficiency.

Therefore, this project proposes the development of an AI-powered personalized learning platform that supports competency assessment exam preparation through intelligent study planning, adaptive practice recommendation, learning analytics, and AI-assisted tutoring.

1.2 Proposed Solutions

The proposed system aims to build an intelligent learning platform that continuously analyzes students' academic performance and automatically personalizes their learning journey throughout the examination preparation process.

The system provides the following capabilities:

- Managing learning materials, question banks, mock examinations, and competency frameworks.
- Conducting diagnostic assessments to identify each student's strengths and weaknesses.
- Generating personalized learning paths according to target scores and available preparation time.
- Recommending adaptive learning content and practice questions.
- Monitoring learning progress using learning analytics dashboards.
- Providing AI-powered tutoring for question explanation, concept clarification, and knowledge summarization.

- Predicting examination performance and recommending learning adjustments.
- Supporting teachers and parents in monitoring students' learning progress through analytical reports.

1.2.1 System Roles

Administrator

- Manage users and system roles.
- Manage courses and competency frameworks.
- Manage question banks and learning resources.
- Configure AI services and system settings.
- Monitor overall platform operations.

Teacher / Academic Manager

- Develop and organize learning materials.
- Manage knowledge domains and learning objectives.
- Create and maintain mock examinations.
- Evaluate question quality.
- Review students' learning progress.
- Monitor AI recommendation quality.

Student

- Complete diagnostic assessments.
- Set target examination scores.
- Follow personalized learning paths.
- Practice adaptive exercises and mock examinations.
- View learning dashboards.
- Interact with AI Tutor.
- Receive personalized study recommendations.
- Track predicted examination performance.

Parent

- Monitor students' learning progress.
- View competency development reports.
- Receive learning alerts and notifications.
- Track target score achievement.
- Monitor study schedules and examination readiness.

AI Recommendation Engine and Chat Bot

- Analyze learning profiles.
- Predict examination performance.
- Explain examination questions.
- Provide detailed answer explanations.
- Answer students' academic questions.
- Generate additional practice questions.

1.2.2 Research-Based Learning Topics

- Adaptive Learning Systems
- Intelligent Tutoring Systems
- Educational Recommendation Systems
- Knowledge Tracing
- Student Performance Prediction
- Learning Analytics
- Educational Data Mining
- Large Language Models for Education
- Retrieval-Augmented Generation

1.3 Functional Requirements

1.3.1 Core Flow 1: Competency Diagnosis and Learning Profile Generation

Students begin by taking a diagnostic assessment designed to evaluate their competency across multiple knowledge domains and cognitive skills.

The system analyzes assessment results together with behavioral learning indicators to establish a comprehensive Learning Profile.

1.3.2 Core Flow 2: Personalized Learning Path Planning

Based on the generated Learning Profile, the student's target examination score, and the available preparation period, the AI engine constructs a personalized study roadmap.

1.3.3 Core Flow 3: Adaptive Learning and Continuous Recommendation

Throughout the learning process, students complete lessons, quizzes, and mock examinations recommended by the system.

1.3.4 Core Flow 4: AI-assisted Learning Support

Students can interact with an AI Tutor at any stage of their learning journey.

1.3.5 Core Flow 5: Learning Analytics and Progress Monitoring

The platform continuously collects learning activities and visualizes these indicators through interactive dashboards.

1.3.6 Core Flow 6: Examination Performance Prediction

The AI engine predicts each student's examination performance based on historical learning data, competency progression, and behavioral analytics.

1.4 Non-Functional Requirements

1. Average response time below 3 seconds for standard operations.
2. Support for concurrent access by at least 1,000 active users.
3. Secure authentication using JWT with Role-Based Access Control (RBAC).
4. Modular and scalable architecture separating Mobile, Web, Backend, AI Services, and Analytics components.
5. Secure storage of learning data with scheduled backup and recovery mechanisms.

6. Cloud-native deployment supporting high availability and horizontal scalability.
7. APIs designed for future integration with Learning Management Systems (LMS) and additional examination frameworks.

1.5 Theory and Practical Technologies

Component	Technology
Mobile Application	Flutter (Android and iOS)
Web Application	ReactJS / Next.js
Backend Services	ASP.NET Core Web API / Node.js Express
Database	PostgreSQL
Authentication	JWT and OAuth2
Large Language Models	LLMs
AI Integration	OpenAI GPT API
RAG	LangChain / Semantic Kernel
Vector Database	Qdrant / Pinecone
Cloud Deployment	Microsoft Azure
CI/CD	GitHub Actions
Version Control	GitHub

Bảng 1: Technology Stack

1.6 Expected Deliverables

1. Mobile application for students.
2. Web portal for teachers and academic administrators.
3. Parent portal for learning progress monitoring.
4. Backend platform for learning data management.
5. AI Recommendation Engine.
6. AI Tutor integrated with LLM and RAG.
7. Learning Analytics Dashboard.
8. Examination Performance Prediction Module.
9. Personalized Learning Path Management Module.

1.7 Proposed Tasks

No.	Task	Member
1	Perform requirement analysis, system architecture design, database design, and backend API development	All
2	Develop the Flutter mobile application for students, including personalized learning, adaptive practice, AI Tutor interaction, and learning progress tracking	1
3	Develop the web portal for administrators, teachers, and parents, including learning content management, question bank administration, dashboards, reports, and user management	2
4	Design and implement AI components, including the Recommendation Engine, Knowledge Tracing, Student Performance Prediction, and LLM-based AI Tutor with Retrieval-Augmented Generation	3
5	Conduct system testing, performance optimization, cloud deployment, technical documentation, and final Capstone report preparation	All

1.8 Other Comments

Additional comments and related proposals will be developed during the implementation and evaluation stages of the project.